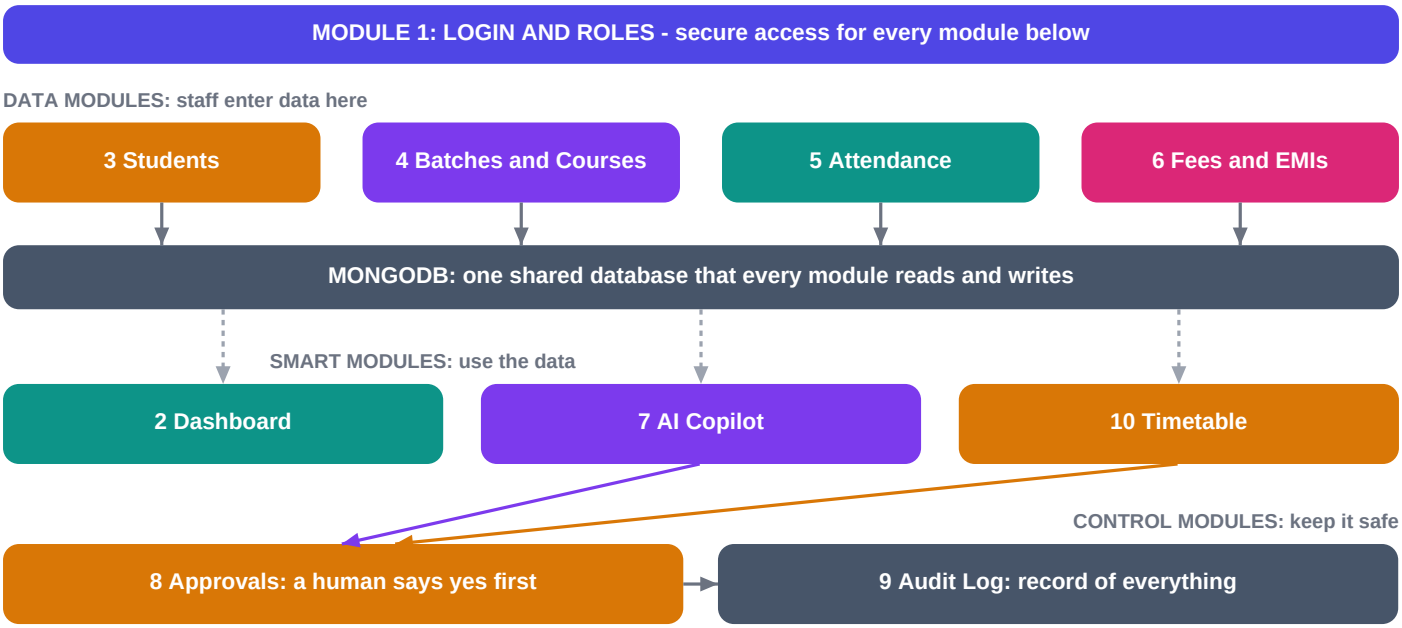
InstituteIQ: All Modules, Screen by Screen

Every module from start to end: what it is, how the screen looks, how it works

The portal has 10 modules. Four of them hold the institute's data, three use that data in smart ways, and two keep everything safe. One login module protects them all. The next pages explain each module in the same way: what it is, who uses it, what is inside, a sample screen, and how it works step by step.



Module list

#	Module	Used by	Main job	Page
1	Login and Roles	Everyone	Sign in, decide which menu and data each role gets	2
2	Dashboard	Owner (mainly)	Show today's numbers and problems	2
3	Students	Counselor, Owner	Master list of students and profiles	3
4	Batches and Courses	Owner	Courses, batches, trainers, rooms, seats	3
5	Attendance	Trainer	Record classroom and online attendance	4
6	Fees and EMIs	Accounts, Owner	Track instalments: paid, pending, overdue	4
7	AI Copilot	All roles (own data)	Answer questions, draft actions, show the query	5
8	Approvals	Owner, Accounts	Human approves every change or message	6
9	Audit Log	Owner	Permanent record of all actions and denials	6
10	Timetable	Owner	Clash-free weekly plan and instant re-plan	7
+	Bonus modules	Owner, Counselor	Dropout risk, payment matching, certificates	8

Who types what? (how data gets in)

Data	Who enters it	How
Students, courses, batches	Counselor or Owner	Add forms (CSV import optional)
Classroom attendance	Trainer	Tick present or absent at class start
Online attendance	Imported	Join log from the online class tool (seeded in demo)
EMI plan	System	Created automatically when a student enrols
EMI payments	Accounts	Mark paid button (statement upload is a bonus)
Trainer availability, rooms	Owner	Settings form or seeded data

Rules that apply to every module

- Every list and screen is filtered by the user's role on the server.
- Every change is saved in the audit log.
- Anything created by the AI waits for human approval.
- Keep the data-entry forms small: seed data does the heavy lifting in the demo.

Modules 1 and 2: Login and Dashboard

The front door and the home screen

MODULE 1 Login and Roles

MUST HAVE

WHAT IT IS

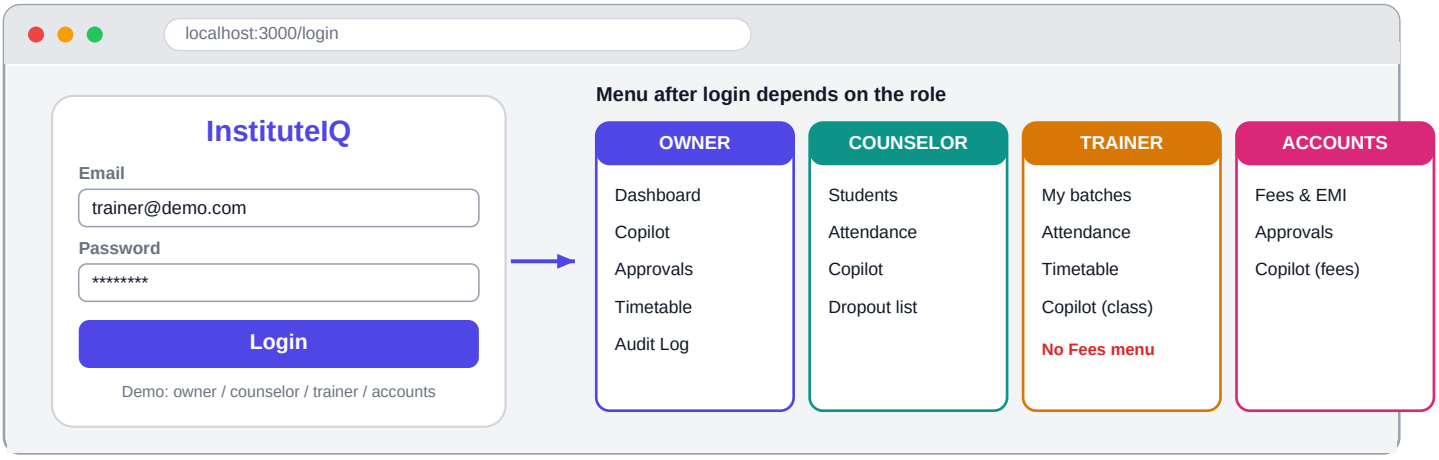
The front door of the portal. Staff sign in with email and password. The server gives back a secure token (JWT) that says who they are and which role they have.

WHO USES IT

Everyone: Owner, Counselor, Trainer and Accounts.

WHAT IS INSIDE

- Login screen with email and password
- Role stored inside the token
- Menu changes by role (Trainer has no Fees menu)
- Server re-checks the role on every request



HOW IT WORKS

1. **Enter login** — Staff types email and password
2. **Server checks** — Password compared with saved hash
3. **Token issued** — JWT holds user id and role
4. **Menu built** — React shows only allowed screens
5. **Every request** — Server checks role before sending data

MODULE 2 Dashboard

SHOULD HAVE

WHAT IT IS

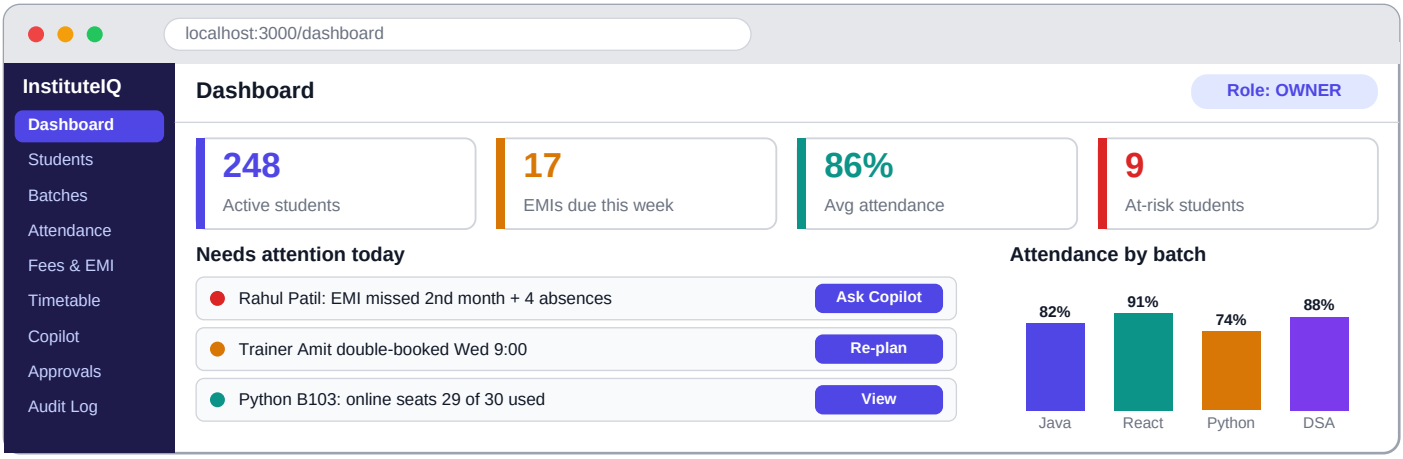
The home screen. It shows the numbers and the problems that need attention today, so nobody has to search sheets for them.

WHO USES IT

Mainly the Owner. Other roles see a smaller version with only their own data.

WHAT IS INSIDE

- 4 number cards: students, EMIs due, attendance, at-risk
- Needs-attention list with quick buttons
- Attendance by batch chart
- Click an alert to jump to the Copilot or Timetable



HOW IT WORKS

1. **Page opens** — React calls GET /api/dashboard
2. **Server counts** — MongoDB counts students, EMIs, absences
3. **Role filter** — Trainer view hides fee numbers
4. **Alerts built** — Rules find missed EMIs and clashes
5. **Click an alert** — Opens Copilot, Timetable or a list

Modules 3 and 4: Students, Batches and Courses

The people and the classes

MODULE 3 Students

MUST HAVE

WHAT IT IS

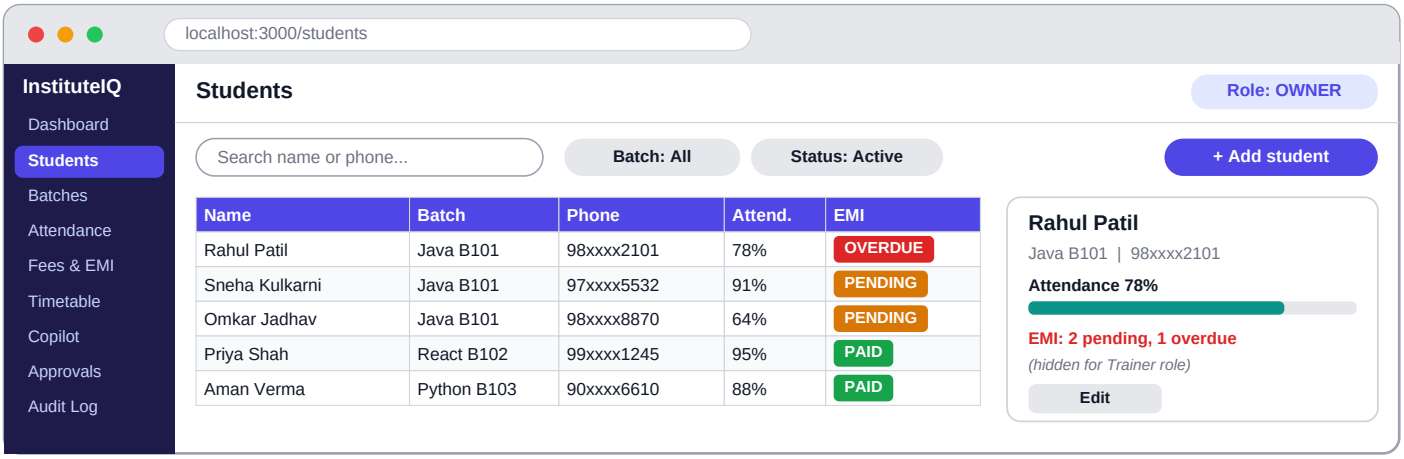
The master list of every student. Each profile links to the student's batch, attendance and EMIs.

WHO USES IT

Counselor and Owner add and edit. Trainer sees own batches only. Accounts sees name and batch.

WHAT IS INSIDE

- Searchable table with filters
- Add student form: name, phone, course, batch
- Profile card with attendance % and EMI status
- Bulk CSV import (optional)



HOW IT WORKS

1. **Add student** — Counselor fills name, phone, batch
2. **Saved** — Stored in students collection
3. **EMI plan made** — System creates the instalments
4. **Lists update** — Batch, attendance, fee screens show it
5. **Search** — Filter by name, batch or status

MODULE 4 Batches and Courses

MUST HAVE

WHAT IT IS

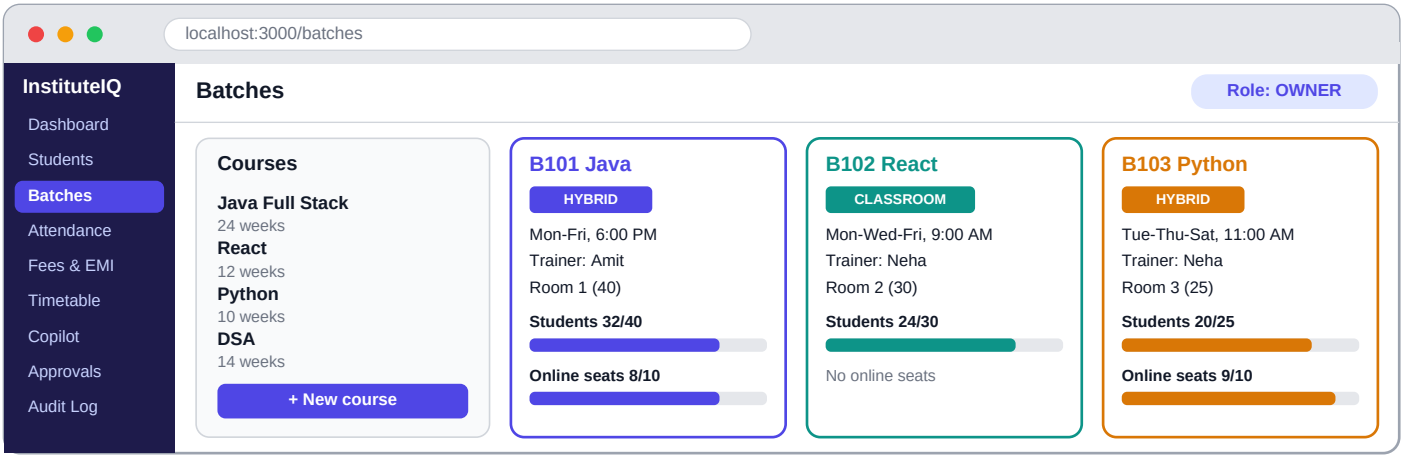
Defines what is taught and when. A course is the subject. A batch is a group learning it with one trainer, room and timing.

WHO USES IT

Owner creates and edits. Everyone can view the batches relevant to them.

WHAT IS INSIDE

- Course list with duration and total fee
- Batch cards: days, time, trainer, room
- Seat usage: classroom and online
- Mode: classroom, online or hybrid



HOW IT WORKS

1. **Create course** — Title, weeks, total fee
2. **Create batch** — Pick course, trainer, days, time
3. **Set seats** — Room capacity and online seats
4. **Assign students** — Counselor puts students in the batch
5. **Feeds timetable** — Scheduler reads these details

Modules 5 and 6: Attendance and Fees

Daily records and money tracking

MODULE 5 Attendance

MUST HAVE

WHAT IT IS

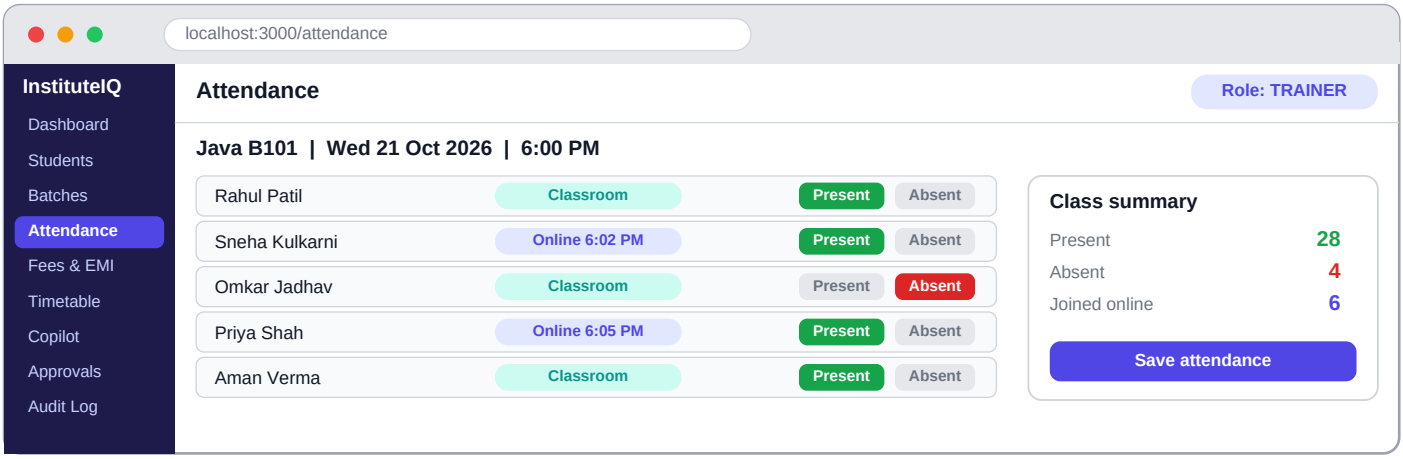
Records who attended each class, in the classroom and online. This data powers the absence alerts and the Copilot's answers.

WHO USES IT

Trainer marks it for own batches. Owner and Counselor view it.

WHAT IS INSIDE

- Mark screen: Present or Absent per student
- Mode tag: classroom or online with join time
- Class summary counts
- Attendance % per student and batch



HOW IT WORKS

1. **Open class** — Trainer picks batch and date
2. **Mark students** — Tick present or absent
3. **Online joins** — Join log fills ONLINE rows (seeded in demo)
4. **Save** — Stored in attendance collection
5. **Used everywhere** — Dashboard and Copilot read it at once

MODULE 6 Fees and EMIs

MUST HAVE

WHAT IT IS

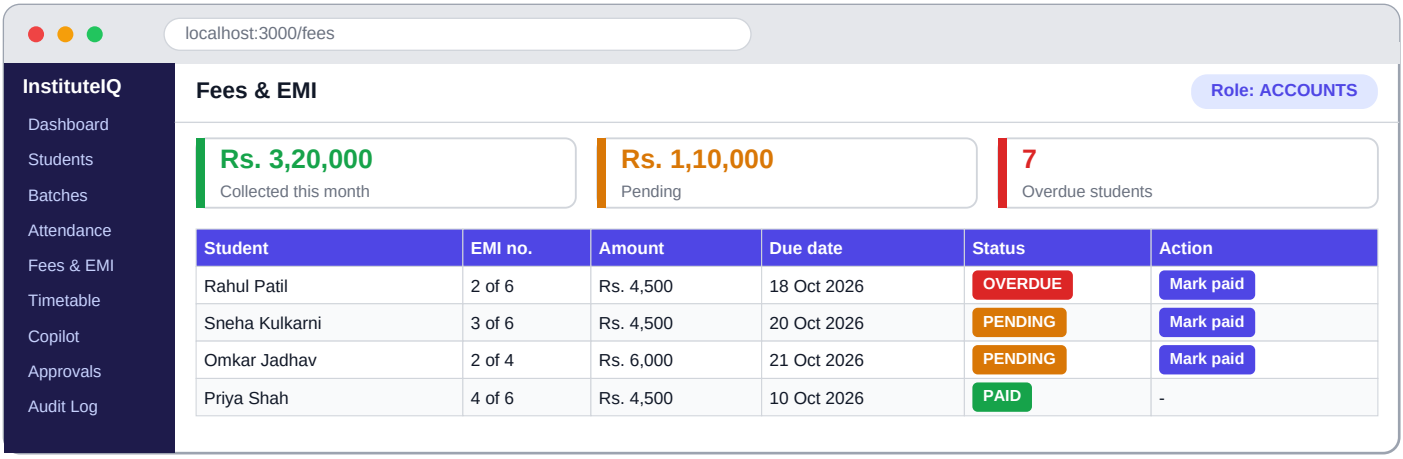
Tracks the course fee split into monthly instalments (EMIs) and whether each one is paid, pending or overdue.

WHO USES IT

Accounts and Owner. Counselor sees status only. Trainer sees nothing (blocked).

WHAT IS INSIDE

- Cards: collected, pending, overdue
- EMI table with coloured status badges
- Mark paid button (saved in the audit log)
- Overdue rows highlighted automatically



HOW IT WORKS

1. **Plan created** — EMIs made when the student enrols
2. **Due date near** — Status stays PENDING
3. **Payment received** — Accounts clicks Mark paid
4. **Missed** — Past due date becomes OVERDUE
5. **Reminder** — Copilot drafts it, human approves

MODULE 7 **AI Copilot** MUST HAVE

WHAT IT IS

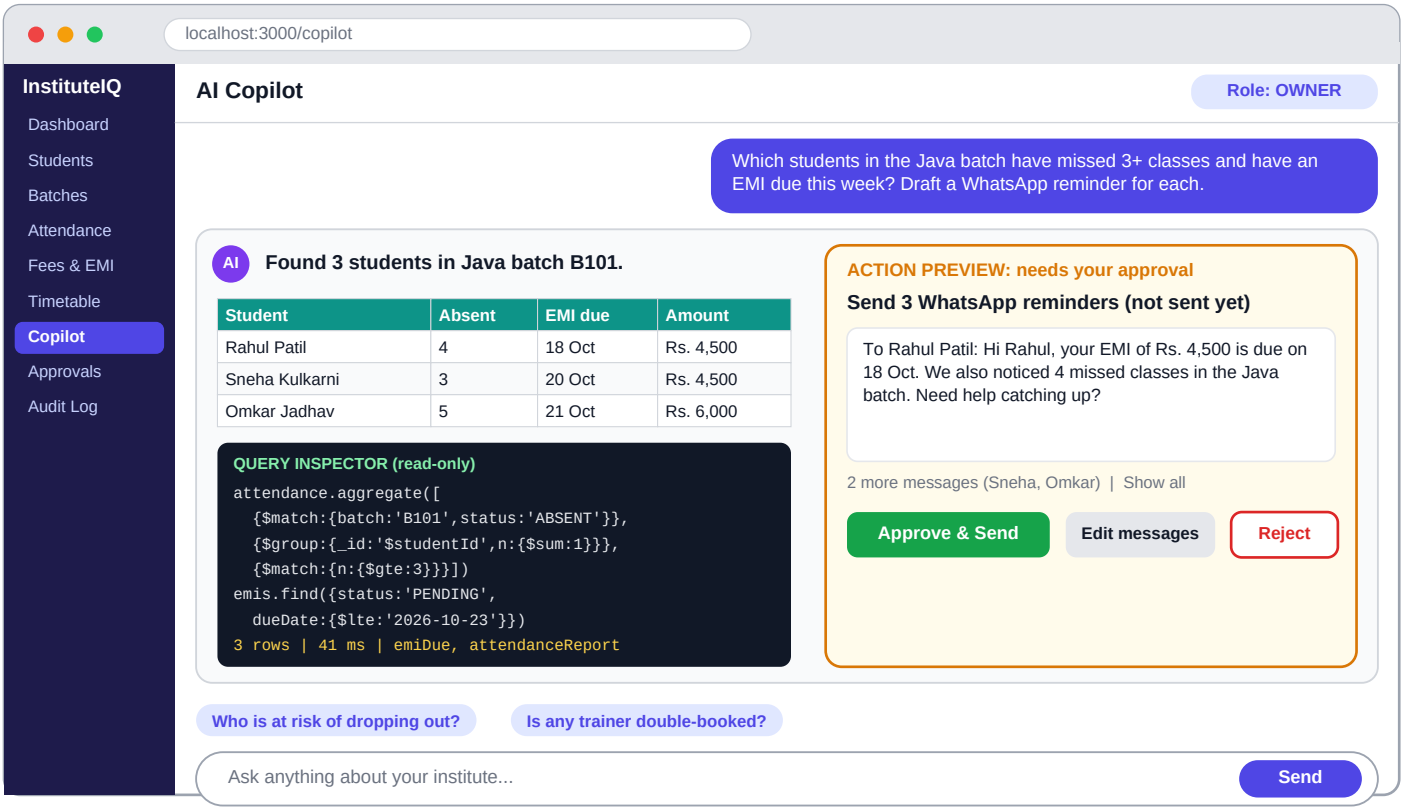
A chat screen where staff ask questions in normal English. The AI agent picks safe tools, reads the data, shows the answer and the exact query, and drafts any action for approval.

WHO USES IT

All four roles, but each gets only the tools and data its role allows.

WHAT IS INSIDE

- Chat with suggestion chips
- Result table with the answer
- Query Inspector: exact query, time, tools used
- Action preview with Approve, Edit, Reject



HOW IT WORKS

1. **You type** — Plain English question
2. **Server checks** — JWT, role, allowed tools
3. **AI picks tools** — Safe, typed arguments only
4. **Tools read data** — Read-only; fee fields masked
5. **Answer + preview** — Table, query, draft action

Example questions by role

Role	Example question	What the Copilot does
Owner	Which batch has the lowest attendance this month?	Reads attendance, shows a ranked table and the query
Counselor	Which students are at risk of dropping out?	Combines absences and EMI status, lists reasons
Trainer	Who was absent in my React batch yesterday?	Shows attendance for own batch only; fee questions refused
Accounts	List overdue EMIs and draft reminders	Reads EMIs, drafts messages, waits in Approvals

Tools behind the Copilot

Read tools: findStudents, attendanceReport, emiDue, feeSummary, batchOverview, trainerSchedule, dropoutRisk. Propose tools (need approval): draftReminder, proposeReschedule, markEmiPaid. Each tool has typed inputs, checks the user's role and records the query it ran.

Modules 8 and 9: Approvals and Audit Log

The safety layer: humans decide and everything is recorded

MODULE 8 Approvals

MUST HAVE

WHAT IT IS

The safety gate. When the Copilot or the scheduler wants to change data or send messages, it creates a pending action here. Nothing happens until a human approves.

WHO USES IT

Owner approves everything. Accounts approves fee actions. Counselor approves own reminders.

WHAT IS INSIDE

- Tabs: Pending, Approved, Rejected
- Exact preview of what will be sent or changed
- Approve, Edit or Reject buttons
- Shows who requested it and when



HOW IT WORKS

1. **Action proposed** — Copilot creates a pending action
2. **Preview shown** — Exact messages or change displayed
3. **Human decides** — Approve, Edit or Reject
4. **Executor runs** — Only approved actions are executed
5. **Logged** — Decision and result go to audit log

MODULE 9 Audit Log

MUST HAVE

WHAT IT IS

A permanent diary of what happened: questions asked, queries run, approvals and denied attempts.

WHO USES IT

Owner sees all. Accounts sees own actions. Other roles cannot open it.

WHAT IS INSIDE

- Time, user, role, event and result per line
- DENIED entries in red (like a trainer asking about fees)
- Filter by user, event or date
- Click a line to see the exact query



HOW IT WORKS

1. **Something happens** — Question, tool call or approval
2. **Entry written** — Server saves user, role, event, query
3. **Never edited** — Entries are only ever added
4. **Owner opens it** — Filters by user, event or date
5. **Proof on demand** — Shows the query behind any answer

Module 10: Timetable

A clash-free weekly plan that re-plans in seconds and explains itself

MODULE 10 **Timetable** — MUST HAVE

WHAT IT IS

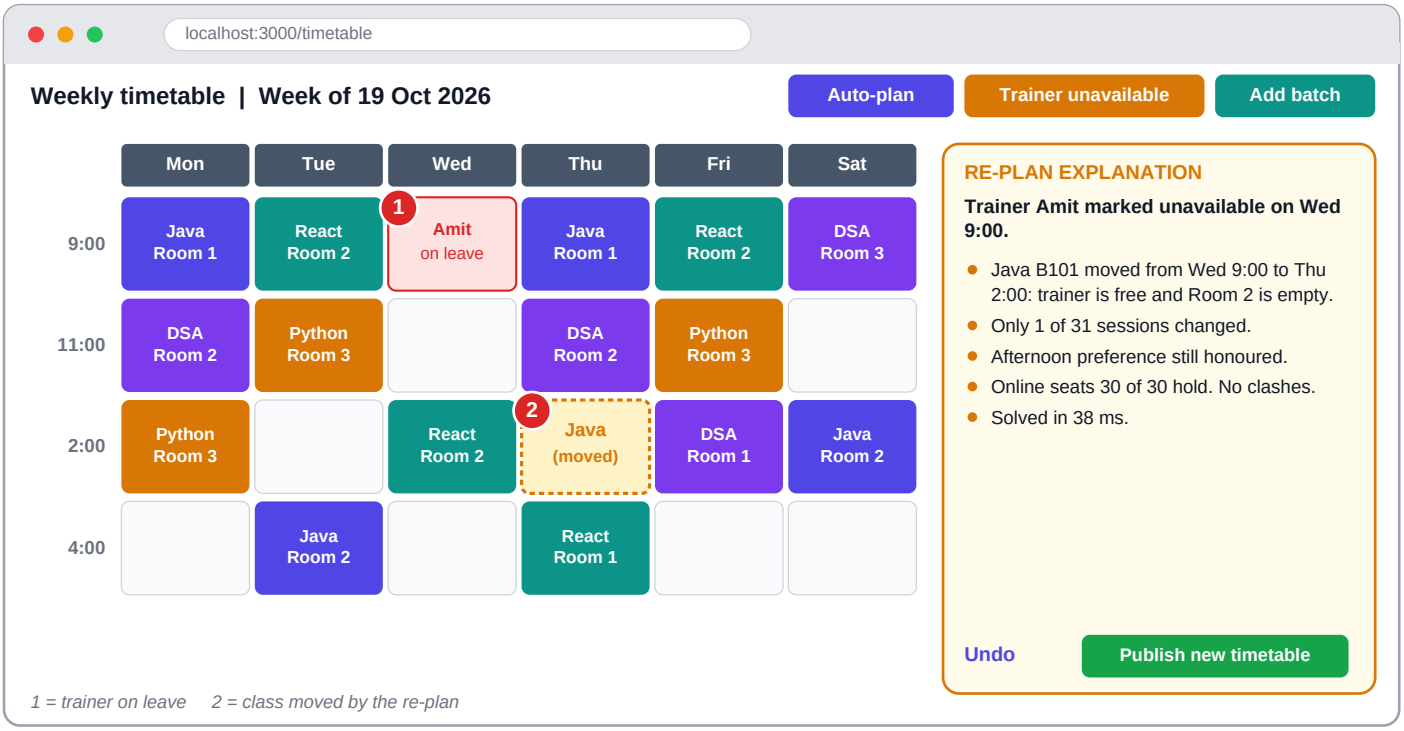
Builds the weekly schedule so no trainer, room or batch is in two places at once. When something changes, it moves only the affected classes and explains each move.

WHO USES IT

Owner creates and re-plans. Counselor views. Trainer sees own.

WHAT IS INSIDE

- Weekly grid: days by time slots
- Auto-plan, Trainer unavailable and Add batch buttons
- Re-plan explanation panel with reasons
- Publish and Undo for the new plan



HOW IT WORKS

1. **Read inputs** — Trainer free times, rooms, batch timings
2. **Auto-plan** — Greedy solver places each batch
3. **Change happens** — Trainer leave or new batch
4. **Re-plan** — Only broken sessions are moved
5. **Explain** — Reason saved for every move

Rules the scheduler follows

- No trainer, room or batch in two places at once.
- Room capacity must be at least the batch size; online seats must not exceed the limit.
- Trainer only teaches in their available hours.
- Batch timing preference (morning or afternoon) is respected where possible.
- Re-plan moves the fewest sessions: same day first, then another day, then a substitute trainer with the skill.

```
function replan(slots, event) {
  const broken = slots.filter(s => clashes(s, event));
  for (const s of broken) {
    const options = candidates(s)
      .filter(o => isFree(o, slots));
    const best = options.sort(byLeastChange)[0];
    moves.push({ from: s, to: best, why: reason(s, best) });
  }
  return { moves, ms: elapsed() };
}
```

Bonus Modules and How Everything Connects

Extra features if time is left, plus the build order

Bonus modules (only after all MUST modules work)

Dropout Risk

Score from attendance, payments and assignments. AI writes the reasons.

Rahul Patil 82

2 EMIs missed, 4 absences

Sneha Kulkarni 55

Attendance fell 20%

Omkar Jadhav 30

Minor: 1 late EMI

Payment Matching

Upload a UPI or bank statement. The system matches payments to expected EMIs.

statement_oct.csv (3 rows)

UPI 4,500 Rahul P. MATCHED

Bank 6,000 O Jadhav MATCHED

UPI 3,000 Unknown REVIEW

Certificates

Create a completion certificate PDF for a student who finished a course.

CERTIFICATE

of Completion

Rahul Patil

has completed Java Full Stack

Demo Institute | Oct 2026

Download PDF

How all modules connect

Staff enter data
Students, Batches, Attendance, Fees

→

MongoDB
One shared database

→

Modules read it
Dashboard, Copilot, Timetable

→

Approvals
Human approves changes

→

Audit Log
Everything recorded

Approved actions write back to MongoDB, so every screen updates. Example: a trainer marks a student absent, and the next Copilot answer and the Dashboard count already include it.

Suggested build order

Module	Build in hours	Effort	Priority
1 Login and Roles	2 - 6	Low	MUST
3 Students, 4 Batches, 5 Attendance, 6 Fees (list and entry screens)	4 - 8	Low-Medium	MUST
7 AI Copilot (agent loop, tools, chat screen)	6 - 16	High	MUST
8 Approvals	12 - 16	Medium	MUST
9 Audit Log and 2 Dashboard	16 - 19	Low-Medium	MUST / SHOULD
10 Timetable and re-plan	19 - 22	High	MUST
Bonus modules	22 - 23.5	Medium	BONUS

Demo path through the modules

Login as Owner → **Dashboard** see alerts → **Copilot** ask Java question → **Approvals** approve 3 messages → **Audit Log** see the record → **Login** as Trainer → **Copilot** fees refused → **Timetable** re-plan